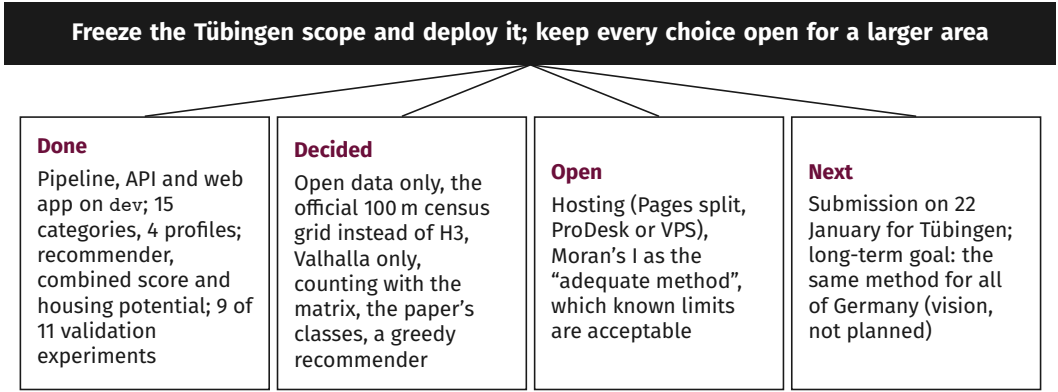
reach: architecture, demo, decisions and the way to all of Germany

Supervisor meeting · DS500 Data Science Project, Topic 9 “The 15-Minute City”

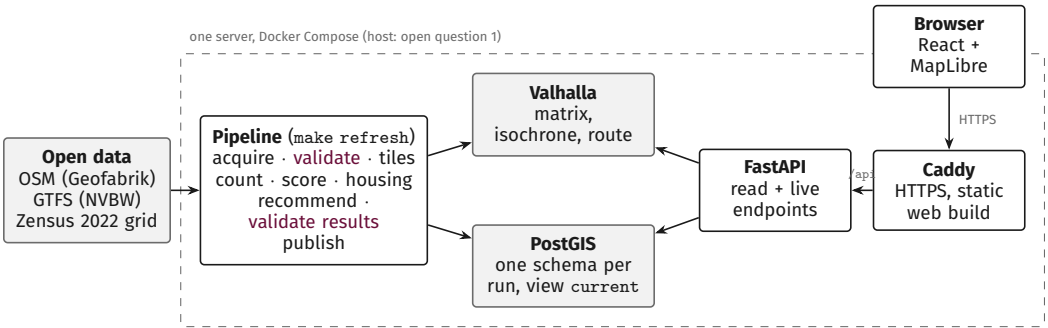
Rico · Socke · Juju
University of Tübingen, winter term 2026/27
12 November 2026

reach runs end to end for Tübingen; we show it live, explain the decisions behind it and ask four questions



Source: run 20261110T233930Z (OSM 8 Nov 2026); validation report (run 20261106T142630Z); docs/architecture.md §13; docs/deployment_proposal.md (proposal, PR #64).

One command turns open data into validated results; the app only reads the published run



Pipeline: one command; a failed check keeps the last good run.

Valhalla only: batch counts and live clicks share one graph.

PostGIS: one schema per run; the API reads only `current`.

Source: docs/architecture.md §2, §3, §7, §8, §13 (O19); weekly timer and production server are planned (§11).

Live demo: the map shows who lacks access, and one click explains why

1,699
populated 100 m cells (90,355 residents)

15 × 4
categories × profiles

What we show, in this order

1. **Class map** per category and profile; categories in six groups
2. **Combined score** 0–100 over chosen categories (display only)
3. **Pin-drop:** isochrone, reachable facilities with times, routes, and the next missing facility beyond 15 minutes
4. **Recommendations:** ranked zones per category with the residents they reach (224 zones)
5. **Housing potential:** the best cells without residents for new homes (830 candidates)
6. **Data page:** sources, snapshot dates and the validation report

Local stack (O16), run 20261110T233930Z; all 15 result checks pass, three warn as known limits.

Source: run 20261110T233930Z (OSM 8 Nov 2026), validate_results report part 2; docs/features.md §1–§5; docs/architecture.md O16 (open).

Every layer of reach rests on a deliberate choice; for each we can name the alternative we rejected

Layer	Our choice	Instead of	Evidence
Data	Open data, one snapshot per run	Live APIs (Overpass)	reproducible
Spatial unit	Official Zensus 2022 100 m grid, cell centroid	H3 hexagons	V6: 1.9 %
Study area	Every resident of the city, 4 km buffer	Built-up area only	N3
Routing	Valhalla, self-hosted, one tile set	OSMnx/NetworkX, r5py	–
Counting	Time matrix; isochrones only drawn	Point-in-polygon	V1: $+18$ %
Profiles	Four profiles, 3.39 km/h default, slopes	One speed, flat	V11
Scoring	k_c, three classes, deficit (paper)	One composite index	–
Recommender	Greedy on the deficit, 100 m zones	Exact solver in every run	V4: ≤ 2.8 %

Why no H3: exact census counts, official cell ids, one grid for all of Germany (backup B1).

Source: docs/architecture.md §2, §4, §6.3, §12, §13; methodology.md §1–§5; notes N3; validation report V1, V4, V6, V11 (run 20261106T142630Z).

Four questions need your answer; for each we bring a recommendation

#	Question	Our recommendation	Blocks
1	Frontend on Cloudflare Pages, backend on one host (proposal of #64, not approved)?	Decide after question 2	Deployment
2	Is the HP ProDesk acceptable as backend host, or is a rented VPS required?	Measure the first full refresh on the ProDesk, then decide	Server, budget
3	Moran's I shows strong clusters on cells (0.72–0.88), none between districts: is this the "adequate method" (M5)?	Yes: optimization plus spatial statistics; LISA map in the app if you wish	Scope of V8
4	Which known limits do you accept (wheelchair on slopes, OSM gaps, extended categories unchecked)?	State them openly; map missing Kitas and schools in OSM	Final report

Also: interim format; database access for you (SSH tunnel?); may we use bwUniCluster; is one city enough, and is a larger area of interest?

Source: docs/deployment_proposal.md (PR #64); methodology.md §7 (M5); validation report V8 (run 20261106T142630Z); notes N11, N13, N16; details in the backup.

By 22 January we finish Tübingen; in the long run the same method can cover all of Germany

Until the submission (planned)

- 25 Nov: interim presentation (`v0.1.0`)
- From ∼1 Dec: server live after the hosting decision
- Dec–Jan: V7 and V9; final categories (M1) and transit classes (M2)
- 22 Jan: submission (`v1.0.0`); 29 Jan: final presentation

All of Germany (vision, not planned)

- Same census file: 3,088,037 populated cells, 82.6 million residents
- Geofabrik extract of Germany; one run per municipality key (AGS)
- Valhalla tiles built on bwUniCluster; vector tiles for the map
- Hard parts: about 1,800× the cells, OSM quality per region, villages need density strata (M3)

Source: docs/features.md §9; project_management_decisions.md §8; Zensus file of run 20261110T233930Z (grid sum, perturbed); architecture.md §11.1, §12 (second region out of scope for Core); methodology.md M1–M3.

Thank you. Your decisions, please.

Hosting, the adequate method and the acceptable limits decide what we freeze next; all of Germany stays our long-term goal.

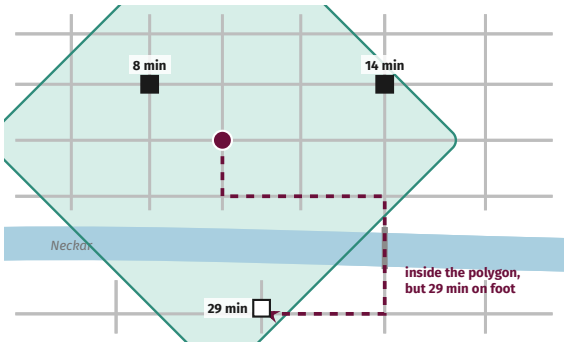
Backup: we keep the official Zensus 100 m grid instead of H3 hexagons

	Zensus 100 m grid (chosen)	H3 hexagons (rejected)
Residents	Exact counts as published	Re-gridding spreads residents by area; errors at every edge
Identity	Official `GITTER_ID` (EPSG:3035) joins other census grid tables (age: M6)	Own index; every join needs re-gridding
Scale	Same grid for all of Germany: 3,088,037 populated cells in the file we already download	Global, but residents must be disaggregated first
Trade-off	Squares: 8 neighbours at two distances	Hexagons: 6 equal neighbours, nicer for spatial statistics

Same grid everywhere: scored cells, recommender candidates, housing cells and the pin-drop snap to it. Origin = cell centroid (O5); with building origins only 1.9 % of residents × categories change class (V6, walking).

Source: docs/architecture.md §2 (principle 4), §5.2, §12 (H3 aggregation out of scope), O5; validation report V6 (run 20261106T142630Z); Zensus file of run 20261110T233930Z.

Backup: counting with isochrone polygons would add 18 % unreachable pairs



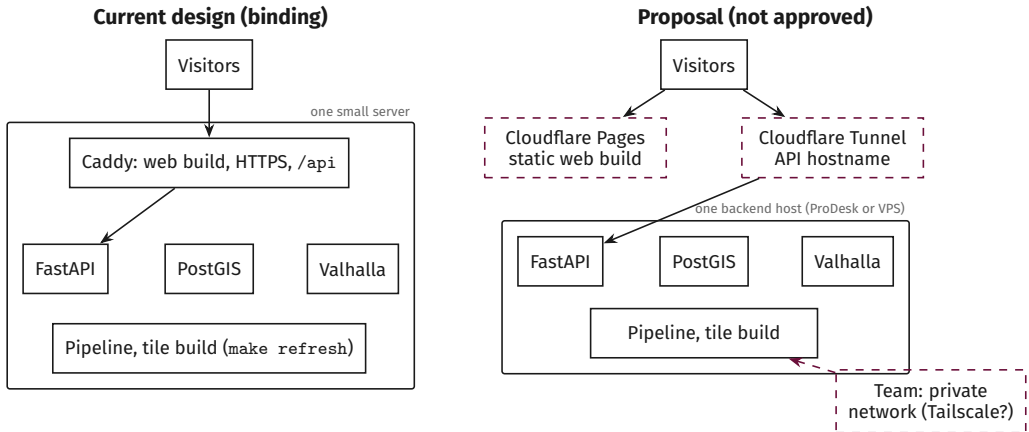
D2: per block of cells one Valhalla matrix to all facilities in reach; a facility counts if its time is $\leq$ 15 min. Isochrones are only drawn.

V1 (150 cells, walking 3.39 km/h): polygons add 18 % unreachable pairs, 47 % of them across the Neckar, rail lines or main roads.

O20: exact prefilter; 0 of 1,200 sampled pairs outside the counts are reachable by `/route`.

Source: docs/architecture.md §6.3 (D2), §6.5, O20; validation report V1 (run 20261106T142630Z); run 20261110T233930Z (`false_negatives`).

Backup: questions 1 and 2, current hosting design vs. the proposal of #64



Source: docs/architecture.md §3, §11 (current); docs/deployment_proposal.md (proposal, PR #64): backend on the HP ProDesk (i5-8500T, 16 GB) or a rented VPS ($\geq$ 8 GB, O1).

Backup: question 4, five known limits and what we propose

Limit	Effect	Proposal
Wheelchair on slopes	Steep sections are not avoided; hillside counts are optimistic (N11)	Accept, state on data page
OSM gaps	13 of 104 Kitas and 2 of 38 schools have no OSM match (V2)	Map them in OSM
Extended categories	No completeness check; monotonicity check at 10 of 1,699 cells, fails above 1 % (N16)	Mark as indicative
Cycling recommender off	12 zones would be added; Valhalla needs 4.0 GiB (W6)	Accept, document
Housing candidates	Land use not fully excluded; buildable not known (N15)	Decide after map check

Shares depend strongly on speed and budget (V5): we always name the profile and budget next to a share.

Source: notes N11, N12, N13, N15, N16; validation report V2, V5 (run 20261106T142630Z); run 20261110T233930Z.

Backup: validation experiments V1–V11

#	Experiment	Key result
V1	Matrix vs. isochrone polygons	Polygons add 18–20 % false pairs, 47 % across barriers
V2	OSM completeness	Childcare recall 88 %, precision 98 %; schools 95 % / 95 %
V3	Recommender back-test	80–98 % of picks within 500 m of a removed facility (only the removed deficit counts)
V4	Greedy vs. exact MILP	At most 2.8 % below the optimum (mean 0.17 %)
V5	Sensitivity (k, speed, budget)	Shares move strongly; district rankings stable (median Spearman 0.78–0.95)
V6	Centroid vs. building origin	1.9 % (walking) / 2.8 % (wheelchair) change class
V7	Local knowledge	Not yet run
V8	Spatial statistics	Moran's I 0.72–0.88 on cells; none between districts
V9	Reproducibility	Not yet run
V10	Matrix vs. single routes	Median $\lvert\Delta\rvert$ 0.5 s; 0 of 900 false negatives
V11	With vs. without slopes	3.3–3.6 % change class, mostly on hillsides

Source: docs/validation_report.md, run 20261106T142630Z (Core categories of that run, parks and playgrounds still one category); the extended categories are not validated yet.